Analisis del Circuito por superposicion

Circuito 1 Apagado F(o)t

$$X(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p = \frac{dFs_1(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - Fs_1(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - Fs_1(t)}{R}$$

$$C_p \frac{dFs_1(t)}{dt} = C_s \frac{d[F(t) - Fs_1(t)]}{dt} + \frac{F(t) - Fs_1(t)}{R}$$

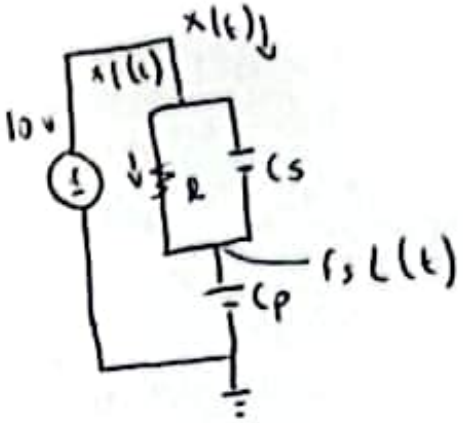
$$C_p s Fs_1(s) = C_s s [F(s) - Fs_1(s)] + \frac{F(s) - Fs_1(s)}{R}$$

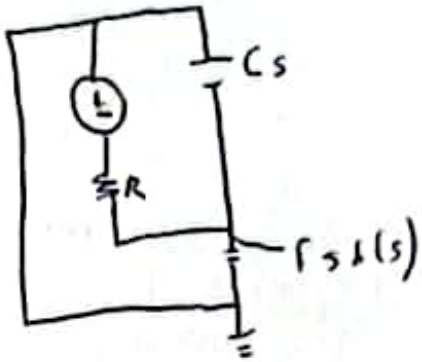
$$C_p s Fs_1(s) = C_s s F(s) - C_s s Fs_1(s) + \frac{F(s)}{R} = \frac{Fs_1 C_s}{R}$$

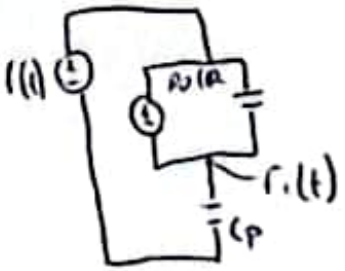
$$\left(C_p s + C_s s + \frac{1}{R}\right) Fs_1(s) = \left(C_s s + \frac{1}{R}\right) F C_s)$$

$$\frac{R(C_p + C_s)s + 1}{R} Fs_1(s) = \left(\frac{C_s R s + 1}{R}\right) F C_s)$$

$$Fs_1(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$







Circuito 2: Apagado f(t)

$$C_p \frac{dfsp(s)}{dt} = C_s \frac{d[x(t) - Fs_2(t)]}{dt} + x(t) - \frac{Fs_2(t)}{R}$$

$$C_p s Fs_2(s) = sS[x(s) - Fs_2(s)] + x(s) - \frac{fs_2(s)}{R}$$

$$C_p s Fs_2(s) = C_s S x(s) - C_s S Fs_2(s) + \frac{x(s)}{R} - \frac{Fs_2(s)}{R}$$

$$\left(C_p s - C_s s + \frac{1}{R}\right) Fs_2(s) = \left(C_s s + \frac{1}{R}\right) t(s)$$

$$\frac{Fs_2(s)}{X(s)} = \frac{C_s s + \frac{1}{R}}{C_p s + C_s s + \frac{1}{R}}$$

$$\frac{Fs_2(s)}{X(s)} = \frac{R(C_s s + 1)}{R(C_p + C_s)s + 1}$$

$$\therefore f(t) = R_s i(t) + \frac{1}{(C_p + C_s)} \int x(t)\,dt$$

$$f_{s2}(t) = \frac{1}{(C_p + C_s)} \int x(t)\,dt$$

$$-x f(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{x(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{f_{s2}(s)}{F(s)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{x(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = \frac{\alpha}{R(C_p + C_s)s + 1}$$

$$f_{s2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Funcion de transferencia

$$f_s(s) = f_{s1}(s) + f_{s2}(s) = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$f_s(s) = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1}$$